

**Volve
15/9-F-12
Maersk Inspirer
USIT/CBL
9 5/8" casing (with 7" tubing elevated)
August 2016
Cement evaluation report**

Title: Volve 15/9-F-12 Maersk Inspirer USIT/CBL 9 5/8" casing (with 7" tubing elevated) August 2016 Cement evaluation report		
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1 General Information

1.1 Well Information

Country	Norway	
Field Name	Volve	
Well Name	15/9-F-12	
Rig Name	Maersk Inspirer	
Well Status during logging	ME deep set plug installed @ 3003 mMD	
Mud Details		
Mud Weight (SG)	1.03	
Mud Type	Brine & sea water mixture	
Clean out run before logging (after cementing)		
Casing Size Logged	9 5/8" (9.625")	Liner (cemented to the top)
From (mMD)	2404	
To (mMD)	3113	
Casing Weight (lb/ft)	53.5	
Nominal ID (in)	8.535	
Inner radius (in)	4.2675	
External radius (in)	4.8125	
Nominal Thickness (in)	0.545	
Previous Casing Shoe Depth (mMD)	14" csg shoe @ 2506 mMD	
Max Deviation in Logged Interval	54°	

1.2 Logging Objectives

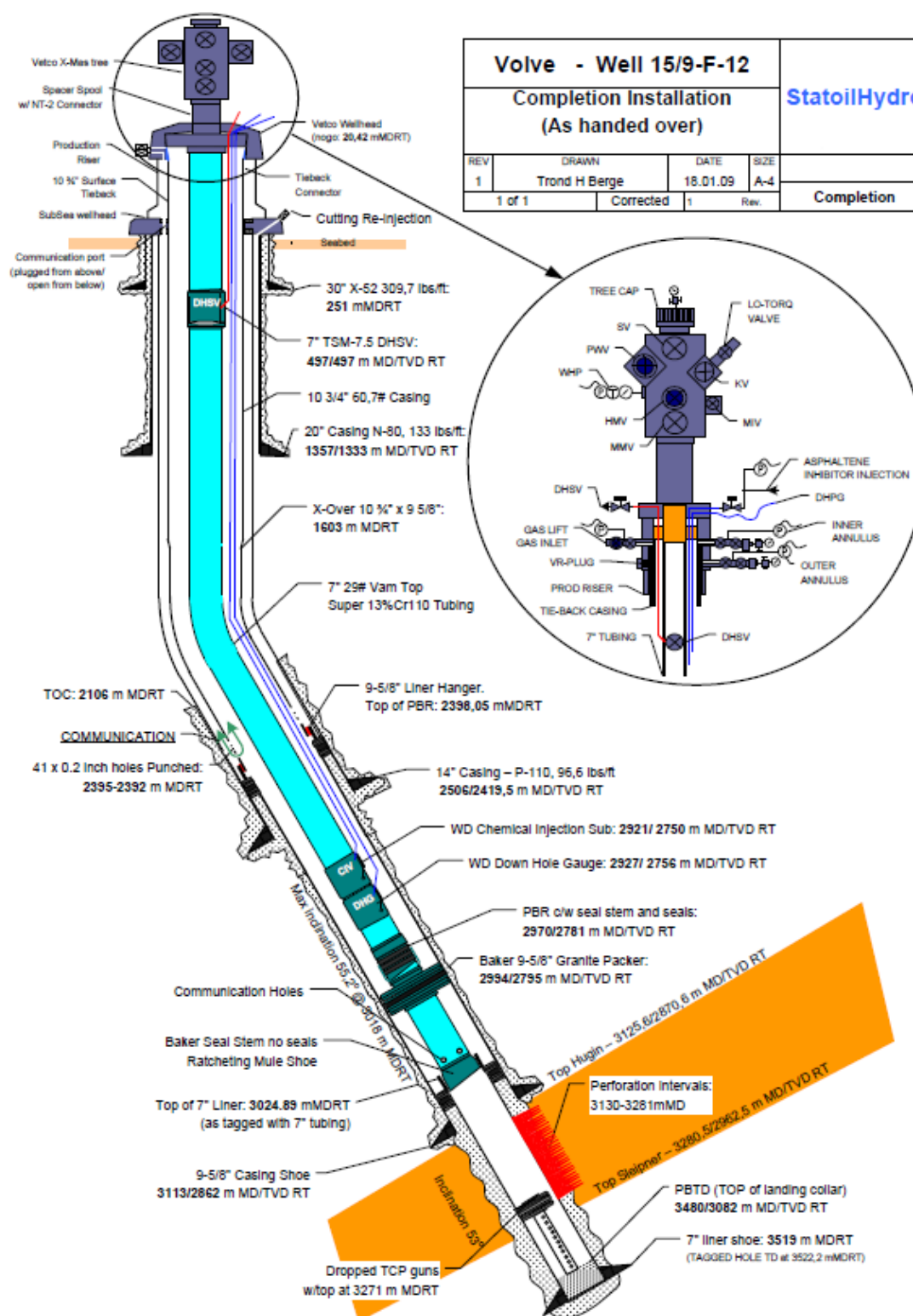
- Verify at least 2 x 30 m cement for primary and secondary barrier in the 9 5/8" liner.

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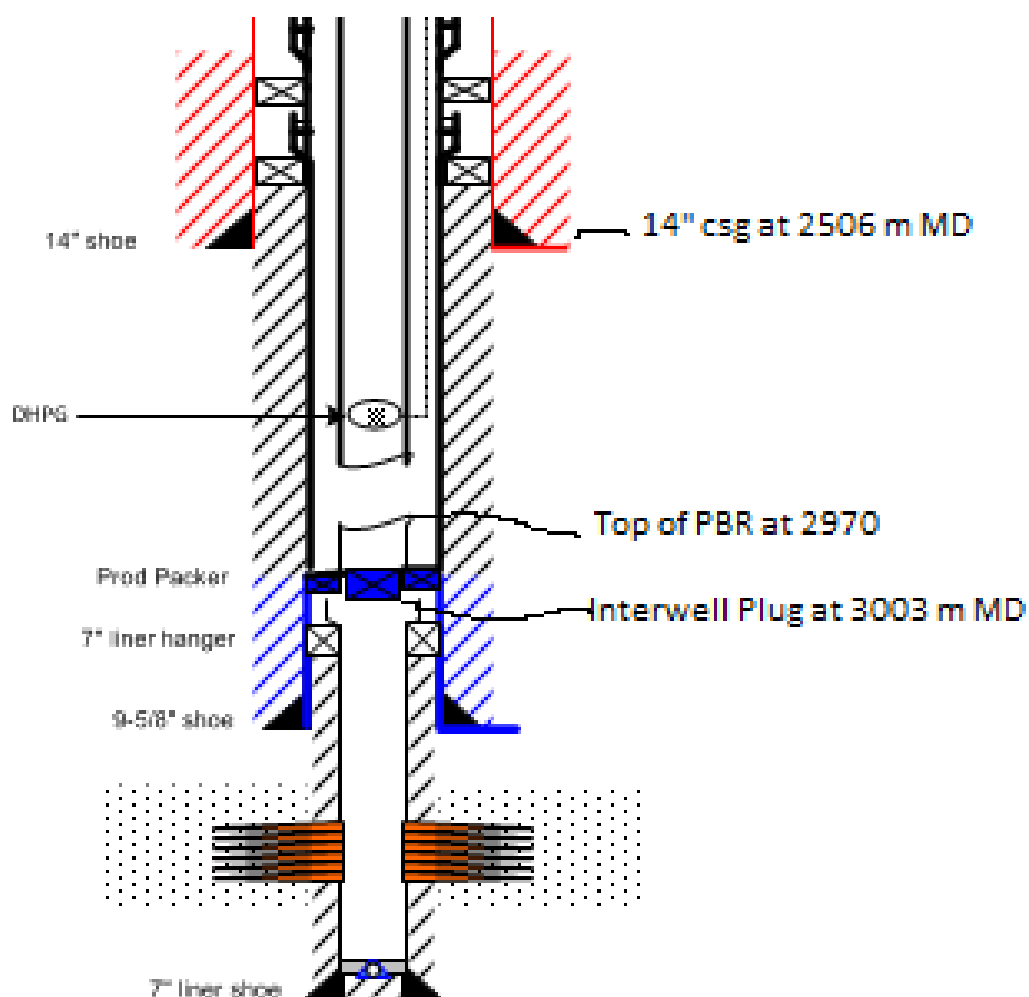


Figure 1: Well sketch (main pass log: 2960–2500 mMD. Tubing end: ~2475 mMD, top PBR: ~2970 mMD)

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1.3 Casing info from DBR

Category/String type			Diameters		RKB hanger m MD	Air gap m MD	From depth m MD	To depth m MD	Date	Description	Tally
Tie-back, Production			10 3/4" x 9 5/8"			54,9	140,1	2 398,1	07.09.2007		
Item type	No. of joints	Diameter inch	Grade	Coupling	Weight lbs/ft	From depth m MD	To depth m MD	To depth m TVD	Remarks		
Casing	123	10 3/4"	13 Cr-80	Vam TOP	60,700	140,1	1 600,1		10 3/4" TIE BACK STRING FROM X-OVER TO HANGER. Hanger hang off at 140,09 mMD. NO of joints incl X-over.		
Casing	67	9 5/8"	13 Cr-80	Vam TOP	53,500	1 600,1	2 398,1		9 5/8" TIE BACK STRING TO X-OVER AT 1600,14 mMD (X-over to 10 3/4" casing). Top pf PBR at 2398,1. Seal stem 2,7 m into PBR.		

Category/String type			Diameters		RKB hanger m MD	Air gap m MD	From depth m MD	To depth m MD	Date	Description	Tally
Liner, Production			9 5/8"		2 413,5	54,9	2 404,4	3 113,0	19.08.2007		
Item type	No. of joints	Diameter inch	Grade	Coupling	Weight lbs/ft	From depth m MD	To depth m MD	To depth m TVD	Remarks		
Casing	60	9 5/8"	13 Cr-80	Vam TOP	53,500	2 404,4	3 113,0	2 865,6	Packer and hanger between 2410.76 m and 2413.49 m. One each spiraglider HD straight per joint last 400 m and on upper joints 54 to 58 having a total number of joints of 60. - Float collar: 3087 mMD		

1.4 Cement Job Information

Cement Job Date and time (when cement displacement was completed)	20 August 2007	
Cementing Details		
Cement Type	Class G	
Cement Density	1.90 SG	
Cement Volume Pumped	29.3 m ³	
Expected Cement Impedance at time of logging	5-6 MRays	(Approximately)
Theoretical Top of Cement	~2135 mMD Meaning the 9 5/8" liner has been cemented "to the top" to above the 14" casing shoe @ 2506 mMD)	Based on actual volume pumped (any cement volume in ratehole not subtracted)
Theoretical Top of Spacer	-	20 m ³ of 1.65 SG WSOBM
Mud weight at the time of cementing	1.40 SG OBM	
Cementing Operations		
Losses during cementing	No	
Any other issues during cementing	No	

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1.4.1 Cementing Report Summary from DBR

Casing/Liner cementing

Wellbore: NO 15/9-F-12

Depth reference is RKB for actual rig and date



Report date: 20.08.2007						
Casing size 9 5/8"	Stage cementing: No		Top up job No	Company Halliburton	Contract no Hall, CEM	
Theoretical TOC MD 2 404,0 m	Evaluated TOC MD m	Evaluated by	Liner rotation planned Yes	Liner rotation achieved		
Remarks						
Fluids pumped	Type	Density g/cm3	Volume m3	Pump rate l/min	Pump pressure bar	Return
Spacer before	Spacer 4 - WSOBM	1,65	20,0	1 783	80,00	Full
Tail	Cement Slurry GTTIZ	1,90	29,3	800	40,00	Full

1.5 Logging Operations

Logging Passes	Date and Time	From (mMD RKB)	To (mMD RKB)	Log Resolution
Free Pipe Pass	No free pipe pass performed, no expected free pipe in 9 5/8"	-	-	-
Repeat Pass	18.08.2016; 18:30-18:40	2960	2855	10° / 6"
Main Pass	18.08.2016; 19:00-19:40	2960	2500	10° / 6"

Tool String	IBC-B / ASLT-B	IBC B-sub to get down passing through 7" tubing (and log in relatively large stand-off).
Tractor Used	No	Tool was run with half springs on the centralizers to overcome friction down, and no gemcos for CBL were applied either.
Pressure applied at surface or WHP when logging If a 'live' well (bar)	No	

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2 Log Quality Control

2.1 Fluid Properties Measurement

Fluid Velocity Range in Main Pass Interval (us/ft)	195-194 us/ft	IBC_FVEL_SEL = Automatic
Mud Impedance calculated using fluid velocity and N=1 (MRayls)	1.61- 1.62 MRayls	Theoretical computation density x velocity (using velocity from above) with N=1
Mud Impedance resulting from free pipe normalisation with N value from MUD_N_FRP applied (MRayls)	n/a	Free pipe normalization was not performed since the 9 5/8" liner had been cemented all the way to the liner top. Additionally, log was run in well known fluid (1.03 SG sea water) with expected n-factor of ~1.0
Mud Impedance values used in final data on which the interpretation is based	1.61-1.62 MRayls	IBC_ZMUD_SEL = Theoretical MUD_N_THE = 1.0

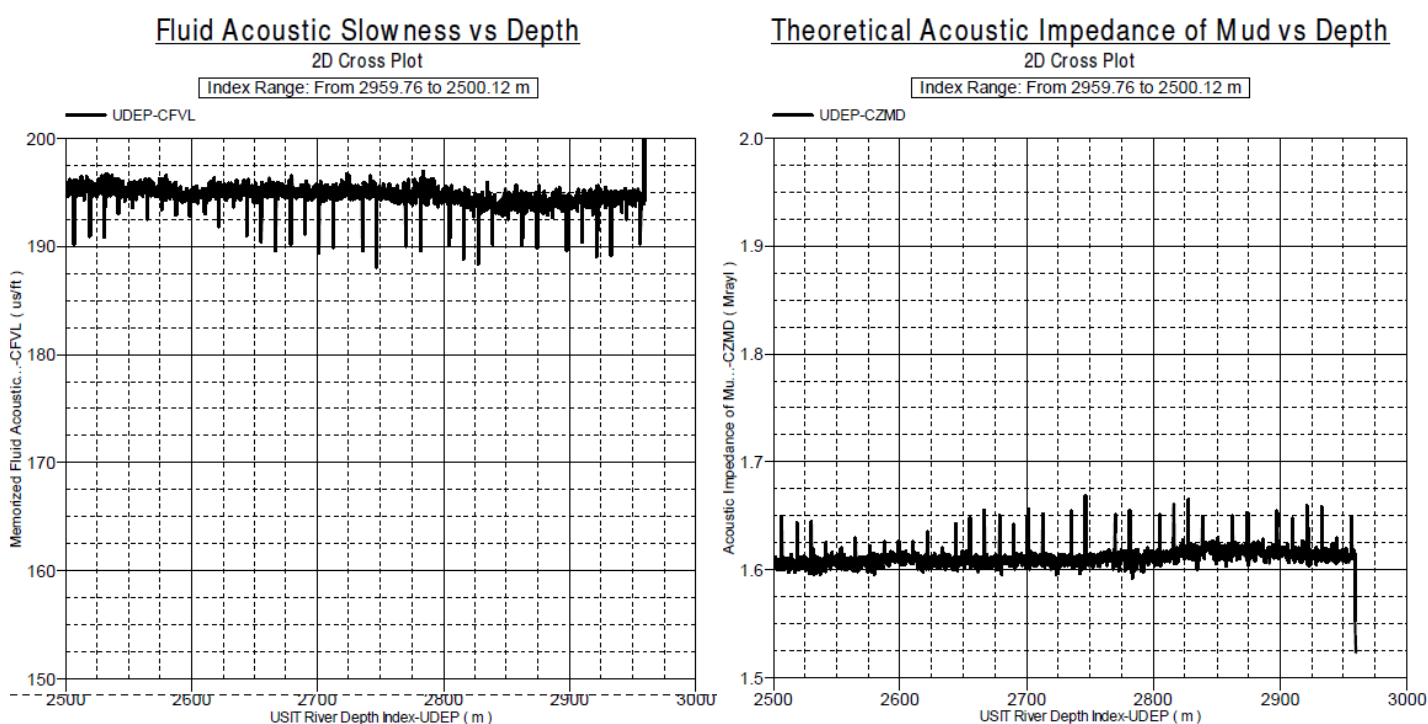


Figure 2: Fluid Properties Crossplots

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2.2 USIT Log Quality

Tool Centralisation	☐	☒	☐	Somewhat eccentralized toolstring, but not considered to have impact on the interpretation.
First Echo Amplitude	☒	☐	☐	
Processing Flags	☒	☐	☐	
Radius Curves within Specification	☒	☐	☐	
Thickness Curves within Specification	☒	☐	☐	
Manufacturing Patterns visible on thickness image	☒	☐	☐	
Quality of Cement Impedance Image	☒	☐	☐	
Repeatability	☒	☐	☐	
Agreement with VDL	☒	☐	☐	

2.3 CBL/VDL Log Quality

Up front of this logging job (CLTE¹), it was expected that the CBL would be very eccentralized and that it could not be used for evaluation.

Transit Time as expected	☐	☒	☐	CBL free pipe adjustment was done inside the 7" tubing @ 685 mMD, with CBAF 0.865. But due to much eccentralization (no gemcos), the CBL curve was not used in the interpretation.
Free Pipe Amplitude as expected	☐	☒	☐	
Tool eccentralisation (transit time variation)	☐	☐	☒	
Repeatability	☐	☒	☐	VDL however, coincided quite well with USIT.

¹ Casing Logging Tubing Elevated.

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3 Interpreted Results

3.1 Interpretation of Annular Contents and Cement/Formation Bond Quality

Interval Top (mMD)	Interval Bottom (mMD)	Annular Contents	Annular Status	Annular Condition Code	Isolating potential based on cement or formation quality (annular distribution and bond quality)[1], [3]	Length of Interval if Isolating Potential is 'High' (NB: see notes [2], [3])	Comments
2500	2505	Cement	Patchy Cement Bond - Low Isolating Potential	PATCem 4B	Low		
2505	2552	Cement	Patchy Cement Bond - Medium Isolating Potential	PATCem 4A	Medium		
2552	2576	Cement	Patchy Cement Bond - Low Isolating Potential	PATCem 4B	Low		
2576	2588	Cement	Well Bonded, heterogeneous, cement around the entire annulus with non-connected small liquid pockets / short liquid filled channels	CEM 1C	High	0 *[2]	See Fig.3 – Interval #1.
2588	2592	Cement	Patchy Cement Bond - Medium Isolating Potential	PATCem 4A	Medium		
2592	2602	Cement	Well Bonded, heterogeneous, cement around the entire annulus with non-connected small liquid pockets / short liquid filled channels	CEM 1C	High	0 *[2]	See Fig.3 – Interval #2.
2602	2607	Cement	Patchy Cement Bond - Medium Isolating Potential	PATCem 4A	Medium		
2607	2625	Cement	Patchy Cement Bond - Low Isolating Potential	PATCem 4B	Low		
2625	2635	Cement	Well Bonded, heterogeneous, cement around the entire annulus	CEM 1B	High	10	46 m cumulative continuous isolating interval. See Fig.3 – Interval #3.
2635	2660	Cement	Well Bonded, heterogeneous, cement around the entire annulus with non-connected small liquid pockets / short liquid filled channels	CEM 1C	High	25	
2660	2671	Cement	Well Bonded, heterogeneous, cement around the entire annulus	CEM 1B	High	11	

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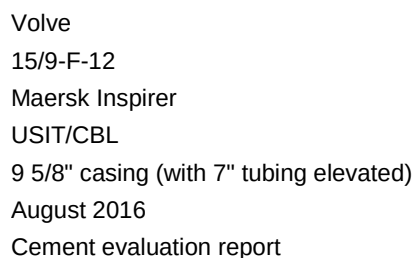
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Interval Top (mMD)	Interval Bottom (mMD)	Annular Contents	Annular Status	Annular Condition Code	Isolating potential based on cement or formation quality (annular distribution and bond quality)[1], [3]	Length of Interval if Isolating Potential is 'High' (NB: see notes [2], [3])	Comments
2671	2684	Cement	Continuous Liquid Filled Channelling – <20% Fluid Channel Width	CHNCem 2A	Low		
2684	2761	Cement	Patchy Cement Bond - Low Isolating Potential	PATCem 4B	Low		
2761	2777	Cement	Patchy Cement Bond - Medium Isolating Potential	PATCem 4A	Medium		
2777	2802	Cement	Patchy Cement Bond - Low Isolating Potential	PATCem 4B	Low		
2802	2807	Cement	Patchy Cement Bond - Medium Isolating Potential	PATCem 4A	Medium		
2807	2848	Cement	Patchy Cement Bond - Low Isolating Potential	PATCem 4B	Low		
2848	2885	Cement	Patchy Cement Bond - Medium Isolating Potential	PATCem 4A	Medium		
2885	2895	Cement	Well Bonded, heterogeneous, cement around the entire annulus with non-connected small liquid pockets / short liquid filled channels	CEM 1C	High	10	75 cumulative continuous isolating interval. See Fig.3 – Interval.#4. (75 m from USIT, 67 m from VDL)
2895	2960	Cement	Well Bonded, heterogeneous, cement around the entire annulus	CEM 1B	High	65	

Notes:

- [1] **High** – Cement or formation material with a log response that satisfies Statoil's requirements such that it can be described as barrier quality. This implies that the interval would be expected to provide isolation, however, for the cement or formation to be considered as a barrier element the cement still needs to satisfy ARIS continuous length requirements – see note [2]
Medium – Implies some sealing potential but with some uncertainty such that it cannot be confirmed with logs alone
Low – Implies no real sealing potential
- [2] ARIS requirements state that a **30m continuous** logged cement interval, interpreted with 'high' isolating potential, is required for it to qualify as a barrier. A dispensation application can be made for **30m cumulative** cement with interpreted 'high' Isolating potential **however** the minimum **continuous** interval that can be considered as contributing to the 30m cumulative total is **10m**.
- [3] Please note that for Formation Bond the current requirements state that a **30m continuous interval** is required for the shale formation to be verified as an annular barrier element.



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Figure 3: Main pass log section

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Appendix B Cement Classification Reference

The Statoil Cement Classification Reference Document can be found here [LINK](#). This document lists all of the standard classification types for the annular material, provides a text description and shows log examples for reference.

Annular Condition Code	Description
CEM 1A	Well Bonded, homogeneous, cement around the entire annulus
CEM 1B	Well Bonded, heterogeneous, cement around the entire annulus
CEM 1C	Well Bonded, heterogeneous, cement around the entire annulus with non-connected small liquid pockets / short liquid filled channels
CHNCem 2A	Continuous Liquid Filled Channelling – <20% Fluid Channel Width
CHNCem 2B	Continuous Liquid Filled Channelling – 20-40% Fluid Channel Width
CHNCem 2C	Continuous Liquid Filled Channelling – >40% Fluid Channel Width
CONCem 3A	Well bonded Cement High Side, Slightly Contaminated Cement Low side
CONCem 3B	Well bonded Cement High Side, Heavily Contaminated Cement Low side
CONCem 3C	Homogeneous Contaminated (or Unset) Cement - Lower than expected impedance
PATCem 4A	Patchy Cement Bond - Medium Isolating potential
PATCem 4B	Patchy Cement Bond - Low Isolating potential
MAWCem 4C	Wet Microannulus – High Isolating Potential
MAWCem 4D	Wet Microannulus – Medium Isolating Potential
MAWCem 4E	Wet Microannulus – Low Isolating Potential
MADCem 4F	Dry Microannulus
GCCem 5	Gas Cut Cement
LLCem 6	Cement in Liner Lap – Eccentralised Liner
FORM 7A	Formation Bond Good /Barrier Quality
FORM 7B	Formation Bond Medium / Not Barrier Quality
FORM 7C	Formation Bond Low/ Not Barrier Quality
FORMCEM7D	Combined Cement and Formation Bond
MUDS 8A	Settled mud solids - High Density / Well Bonded
MUDS 8B	Settled mud solids - Medium Density / Medium Bond
MUDS 8C	Settled mud solids - Low Density / Patchy Bonded
FPL 9A	Liquid Filled Free Pipe
FPG 9B	Gas (or light oil) filled free pipe
OTHER	If the annulus status cannot be classified by one of the above categories
PDQ	Data Quality Issues – such that data is not interpretable

By Default: **Green = High Isolating Potential** / **Orange = Medium Isolating Potential** / **Red = Low Isolating Potential**
 (Isolating potential of the 'Other' category will be decided by the interpreter and is not default)

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Appendix D Logging Sequence of Events

Client: Statoil
Field: Volve
Well: 15/9-F-12

Engineer Day: Luis Aguayo	Operator Day: Per Thorsen
Engineer Night: Sverre Grevsjøtt	Operator Night: Jostein Aga

Run 3

18-Aug-2016

14:20 TBT for Rig up
14:35 Rig up. Stick up length=1.5m
16:00 Zero at head=16.4m. Surface checks.
16:30 RIH
17:00 CBL Free Pipe Adjustment
Tension pick up at 700m= 1200lbs
17:30 Tension pick up at 1200m= 3060lbs
17:40 Tension pick up at 2000m= 3740lbs
17:50 Tension pick up at 2000m= 4540lbs
18:15 Reached TD
18:30 Start Repeat Pass
18:40 Finished repeat Pass
18:50 RIH to 2965mMD
19:00 Start main pass
19:40 finish main pass, process data and upload
19:55 Waiting for vakt to approve data QC
20:10 Ok to POOH
20:12 Stuck at entry to tubing, try at different speeds
20:35 open top centralizer to help tool in to tubing
20:40 Got through
20:40 at surface
20:45 toolbox talk before RD
21:00 start RD
22:15 Finished RD, tools away from drillfloor